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May 10, 1953

FEDERAL SPECIFICATION

IMPRESSION MATERIAL, DENTAL HYDROCOLLOID, ALGINATE TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental alginate impression material, in powder form, suitable for taking oral impressions.

1.2 Classification. Alginate material shall be of the following types as specified (see 6.1):

Type I - Fast set.
Type II - Regular (Normal) set.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

U-S-746 - Stone, Artificial, Dental.
RR-S-366 - Sieves, Standard for Testing Purposes.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

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Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Dental Association (ADA)

ADA Specification No. 18 Alginate Impression Material

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, Illinois 60611.)

3. REQUIREMENTS

3.1 Material. The impression material shall contain alginate as a gel-forming ingredient and shall be uniform in texture and free of foreign matter. When prepared in accordance with instructions (3.2.3) accompanying the package, the material shall form a smooth plastic mass suitable for taking impressions in the mouth. The material shall be furnished clean and ready for use.

3.1.1 Odor and flavor. The material shall have no unpleasant odor or flavor when tested as specified in 4.4.3.

3.1.2 Irritation. The manufacturer shall furnish a statement to the procuring activity (see 6.1), stating that the material does not normally irritate oral tissues and does not contain poisonous ingredients in sufficient concentration to be harmful to human beings when used as directed, or in the event of the accidental ingestion of 10 milliliters (ml).

3.2 Description. Alginate material shall be furnished in the type specified (see 6.1). Type I alginate shall be a fast setting material and type II alginate shall be a regular setting material.

3.2.1 Amount of material in unit container. The volume of material in unit container, when mixed in the proportions recommended by the manufacturer and allowed to set in a suitable mass, shall be sufficient to make a minimum of thirty impressions when tested as specified in 4.4.4.

3.2.2 Accessories. A powder proportioner, suitable for scooping the alginate powder, and a water proportioner shall be furnished in a suitable envelope or container and shall be enclosed in unit container. The proportioners shall be made of suitable rigid plastic. The powder proportioner

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shall accommodate one scoop of powder. The water proportioner shall accommodate between 40 and 60 cubic centimeters (cc) of water. The water proportioner shall be graduated to show the water levels necessary for use in either a "two-scoop system" (manufacturer's instructions describe a one-scoop impression and a two-scoop impression) or a "three-scoop system' (manufacturer's instructions describe a one-scoop impression, a two-scoop impression, and a three-scoop impression). The powder and water proportioners shall be suitable for use in mixing the following types of impressions:

Two-scoop system

Powder	Water	Impression
One scoop	Half measure	Single partial lower or upper impression
Two scoop	Full measure	Full upper or lower impression

Three-scoop system

Powder	Water	Impression
One scoop	1/3 measure	Single partial lower or upper impression
Two scoop	2/3 measure	Single full lower or medium upper impression
Three scoop	Full measure	Large full upper impression

3.2.3 Instructions for use. Adequate and accurate instructions for manipulating the material shall accompany each unit container. Instructions furnished shall be pertinent only to the type of material furnished. In addition, these instructions shall include the following minimum requirements:

1. Temperature of water to be used for making mix: Instructions shall specify the water temperature to be used to obtain a 73 degree Fahrenheit (° F) mix temperature, and the water temperature to be used to compensate for extremes in ambient temperature at 65°, 73°, 75°, 80°, 85°, and 90°F.

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2. Powder-water ratio (in terms of number of scoops to volume of water, and also in grams of powder and milliliters of water).

3. Total working time (i.e. mixing, loading and seating), and the individual time for mixing, loading and seating in the mouth.

4. Initial setting time, (i.e. loss of stickiness to a dry finger, timed from start of mix), the term "FAST SETTING" on unit container for type I material.

5. Withdrawal time (i.e. time in the mouth for proper set and safe withdrawal from mouth, timed from start of mix).

6. Treatment of the impression in the interval between withdrawal from the mouth and preparation of gypsum casts.

7. Method of conditioning the powder prior to scooping, such as rotating or shaking the unit container or drawing the scoop through the powder.

8. Statement indicating method of preparing the patient's mouth prior to introducing the impression material.

Instructions for use shall be lithographed on the exterior of the unit container or shall be furnished on a separate sheet of instructions included in the unit container. General instructions, which are applicable to both types of material, may appear on the exterior of the unit container providing that the information is clearly defined and the uses adequately differentiated with regard to type I and type II material. Any additional instructions which are required (paragraph 3.2.3, subparagraphs 1 through 8), and which do not appear on the unit container shall be enclosed as a separate sheet and shall be included in the container carrying the powder and water accessories. A separate pressure sensitive, color-coded label, with the term "FAST SETTING" may be used to highlight the fast setting material (type I) from the regular (type II) material.

Unless otherwise specified, prior to delivery two copies of the instructions shall be furnished to the Directorate of Medical Materiel, Technical Services Branch, (Code ATTS-2), Defense Personnel Support Center, 2800 S. 20th St., Philadelphia, PA 19101 for information purposes.

3.2.4 Reproducibility. The alginate material, when prepared in accordance with manufacturer's directions, shall impart a smooth, non-chalky surface and separate cleanly from a hardened gypsum cast when tested as specified in 4.4.5. The alginate shall be of such formula as to be compatible with both type I, high density and type II, low density hardened dental stone of U-S-746 (see 6.2).

3.3 Physical properties.

3.3.1 Fineness. All of the material, when tested as specified in 4.4.6, shall pass a number 20 sieve. The sieve shall conform to RR-S-366.

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3.3.2 Mixing time. The time of mixing the impression material in accordance with manufacturer's directions to obtain a smooth workable consistency shall be not more than one minute when tested as specified in 4.4.7. The mixed material shall show no segregation, shall be homogeneous, free from lumps and granules, and shall have a smooth surface, when tested as specified in 4.4.7.

3.3.3 Working time. The working time shall be not less than 1-1/4 minutes for type I material and not less than 2 minutes for type II material, when tested as specified in 4.4.8.

3.3.4 Setting time. Type I material shall have an initial setting time of 1-1/2 to 2 minutes, and type II material shall have an initial setting time of 2-1/2 to 3 minutes, when tested as specified in 4.4.9.

3.3.5 Withdrawal properties. The impression material shall have the following properties: type I material shall have a withdrawal time (finished impression removed from mouth) of not more than 3 minutes, and type II material shall have a withdrawal time (finished impression removed from mouth) of not more than 4 minutes when tested as specified in 4.4.10.

3.3.6 Compatibility with gypsum. A gypsum cast poured against the alginate shall reproduce line B (0.050 millimeters (mm) wide) and line C (0.075 mm wide) of figure 1 when tested as specified in 4.4.11.
Ten minutes after separating the cast from the impression, line C shall not be obliterated when tested in accordance with 4.4.11.

3.3.7 Permanent deformation. The permanent deformation shall be no more than 2.25 percent after 12 percent strain is applied for 30 seconds, when tested as specified in 4.4.12.3.

3.3.8 Compressive strength. The compressive strength shall be no less than 5000 grams per square centimeter (g per sq cm), when tested as specified in 4.4.12.4.

3.3.9 Strain in compression. The strain shall be no less than 10.0 nor more than 20.0 percent between a stress of 100 g per sq cm and a stress of 1000 g per sq cm when tested as specified in 4.4.12.5.

3.3.10 Deterioration. The compressive strength of the specimens, when tested as specified in 4.4.12.4, after storage as described in 4.4.12.6, shall be no less than 4000 g per sq cm. The setting time after storage, shall not change from the original initial setting time (3.3.4) by more than plus or minus 30 seconds, when tested as specified in 4.4.12.6.

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3.3.11 Consistency. The alginate shall not run out of a metal impression tray when tested as specified in 4.4.13.

3.4 Stability. The shelf life of the impression material shall be 24 months from date of manufacture, and no less than 20 months shall remain from date of delivery to the Government when stored at 59° to 86° F. When tested after 24 months from date of manufacture, the compressive strength shall be no less than 4000 g per sq cm and the setting time shall not vary by more than 30 seconds from the original requirement (3.3.4) for the applicable type.

3.5 Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the impression material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 18 for alginate impression material. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

3.6 Lot or batch number. Each unit container shall be marked with a lot or batch number and date of manufacture.

3.7 Workmanship. Workmanship shall be first class throughout. The alginate material shall be free from defects which detract from its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, III, and IV (see 4.2.1, 4.3, and 4.3.1) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one can of impression material (unit of issue).

TABLE I. Inspection and acceptable quality levels.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.5
Minor	S-1	2.5

4.2.2 Sampling for tests. The unit of product for testing shall be one container of powder taken at random from each lot or batch number. Where feasible, the same test unit may be used for a determination of two or more different test characteristics. The number of determinations for each characteristic shall be as listed in table II.

TABLE II. Tests.

Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Odor and flavor	3.1.1	4.4.3	1	1 1/	1
Amount of material in unit container	3.2.1	4.4.4	2	1 1/	2
Reproducibility	3.2.4	4.4.5	1	1 1/	1
Fineness	3.3.1	4.4.6	1	1 1/	1
Mixing time	3.3.2	4.4.7	1	1 1/	1
Working time	3.3.3	4.4.8	1	1 1/	1
Setting time	3.3.4	4.4.9	3	3 2/	9
Withdrawal properties	3.3.5	4.4.10	1	1 1/	1
Compatibility with gypsum	3.3.6	4.4.11	1	1 1/	1

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TABLE II. Tests (cont'd)

Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product.	Total number of determinations
Permanent deformation	3.3.7	4.4.12.3	1	3 2/	3
Compressive strength	3.3.8	4.4.12.4	1	3 2/	3
Strain in compression	3.3.9	4.4.12.5	1	3 2/	3
Deterioration	3.3.10	4.4.12.6	1	3 2/	6 3/
Consistency	3.3.11	4.4.13	1	1 1/	1
Leakage of unit package (can)	5.1.2.1	PPP-C-96	1	1 1/	1

1/ A lot shall be rejected if any determination fails the test requirement.
2/ A lot shall be rejected if the average of the determinations per unit of product fails the test.
3/ Three determinations for compressive strength and three determinations for setting time after deterioration aging.

4.3 Examination. Impression material shall be examined for defects, including, but not limited to, those in table III.

TABLE III. Classification of Defects.

Categories	Defects
Major:	
101	Type not as specified.
1/ 102	Material not free of foreign matter.
Minor:	
1/ 201	Instructions not present with unit container, incorrect, incomplete, or not applicable to material furnished.
1/ 202	Accessories (powder proportioner and water proportioner) not present in unit container in envelope or container.
1/ 203	Powder and water proportioners furnished not of proper size or proportion for the applicable 2 or 3-scoop system outlined in instructions.
1/ 204	Plastic proportioners not rigid.

1/ In process examination may be used for examination for these defects.

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4.3.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE IV. Examination of preparation for delivery.

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging & packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4 Tests.

4.4.1 Preparation of specimens. The specimens shall be prepared in accordance with the manufacturer's directions as furnished with the unit container (3.2.3).

4.4.2 Standard testing conditions, All physical tests shall be made under uniform atmospheric conditions 23.0 ± 2.0 degrees Centigrade (° C) and 50 ± 10 percent relative humidity, unless otherwise specified. For determining all physical tests requiring a mix, the water temperature shall be first adjusted to compensate for the ambient temperature to give a mix of 23.0 ± 0.2° C. This temperature shall be determined by placing an accurate thermometer in the center of the mass in the mixing bowl. Test mixes requiring preparation in accordance with manufacturer's instructions shall have the water temperature adjusted prior to mixing.

4.4.3 Odor and flavor. The pleasantness or unpleasantness of the odor and flavor shall be determined by smelling and tasting the material.

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4.4.4 Amount of material in unit container. Two scoops of powder shall be mixed with water, as recommended by the manufacturer, and the volume recorded to the nearest milliliter. The number of impressions obtainable from the unit container on a 46 ml of mix basis shall be calculated as follows:

$$\text{number of impressions} = \frac{\text{ml of mix (2 scoop)}}{46} \times \frac{\text{g of powder (unit container)}}{\text{g of powder (2 scoops)}}.$$

4.4.5 Reproducibility. A mix of the alginate material shall be made in accordance with the manufacturer's instructions and a clinical impression shall be taken of the maxillary jaw. Using the manufacturer's procedure recommended for preparing the impression for the cast, pour artificial stone (in compliance with U-S-746) against the impression and examine for the cast properties specified in 3.2.4.

4.4.6 Fineness. The container of alginate shall be tumbled or rotated to insure homogeneity of the contents. Ten to twelve grams of the alginate powder as received from the manufacturer shall be placed on a number 20 sieve, the top and bottom shall be replaced, and the assembly shall be vibrated mechanically in a Ro-tap for 10 minutes. On completion of the test, the sieve shall be carefully examined for residue remaining on the upper surface of the wire or occluding the sieve openings.

4.4.7 Mixing time and uniformity. After spatulating impression material in accordance with instructions furnished (for not more than one minute), mix shall be observed for homogeneity, smoothness, and freedom from lumps and granules.

4.4.8 Working time. A penetrometer, using a cylindrical point of 4 mm diameter and a load of 50 g shall be used for this test. A smooth flat plate shall be placed under the point and a fiducial reading made. A ring, 16 mm high and approximately 3 cm inside diameter, shall be placed upon the base plate beneath the point and filled with material mixed in accordance with the manufacturer's instructions. The top surface shall be leveled. Two minutes from the start of mix for type II material, or 1-1/4 minutes for type 1 material, the point shall be placed in contact with the top surface of the material and released. Ten seconds after release, an indicator reading shall be made. The difference between the two readings shall be no more than 0.25 mm. All readings shall be made to the nearest 0.02 mm.

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4.4.9 Setting time. Setting time shall be measured by filling a polymethyl methacrylate or metal ring, 3 cm inside diameter and 16 mm high, or a metal tray approximately 100 by 28 by 20 mm with a mix of alginate material. A scoop portion of alginate powder and the necessary volume of water, mixed in accordance with the instructions furnished by the manufacturer, shall be used for preparing the initial setting time. As the material sets, a polished polymethyl methacrylate solid rod approximately 3/8 inch in diameter shall be inserted vertically into the mixed material, touching the bottom of the tray and shall immediately be withdrawn. The initial setting determination shall be run at 15 second intervals, for both type I and type II material, starting with 2 minutes and 15 seconds for type II and 1 minute and 15 seconds for type I material until the end of the rod cleanly separates from the impression materials. The setting time shall be the time elapsed from start of mix until the alginate fails to adhere to the surface of the rod. Three determinations shall be made and the average shall be reported as the initial setting time.

4.4.10 Withdrawal test. A penetrometer, with a 1.5 mm ball point needle and an arm weight of 50 gm shall be used for this test. A mix of the alginate material, made in accordance with the manufacturer's instructions, shall be placed in a Petri dish on a penetrometer plate. The initial setting time shall be determined by noting loss of stickiness to a dry finger, timed from the start of the mix and tested at 5 second intervals. As soon as the mix sets, the needle shall be adjusted to a firm contact with the top surface of the set alginate which shall be the zero reference on the indicator. Fifteen seconds after the initial set, and each fifteen seconds thereafter, the needle shall be released for five seconds, and the penetration determined by means of the indicator reading. After each test, the needle shall be returned to the zero reference and the Petri dish moved so that a new spot will be tested. This test shall be continued at fifteen second intervals until the needle drops no more than 1.5 mm. The time at which this occurs, measured from the start of mix, shall be the withdrawal time.

4.4.11 Compatibility with gypsum. A mix of the material shall be made in accordance with the manufacturer's instructions. A ring of the type specified in 4.4.12.1 shall be positioned on a stainless steel test block similar to that shown in figure 1, so that the intersection of a crossline and line A, 0.025 mm wide, is in the center of the ring. As an alternate, a ring of the type specified in 4.4.12.1 shall be positioned on a stainless steel test block similar to that shown in figure 1 so that the three lines of the test block are centered on the ring. The stainless steel test blocks may be lightly dusted with talcum powder and the excess talcum powder blown off, if the impression material adheres to the blocks. The ring shall be slightly overfilled with prepared alginate. A flat plate shall be placed on top and the excess material shall be squeezed out. Two minutes after the start of mix, the entire assembly shall be placed in a water bath maintained at 37 $\pm$ 1°C. Six minutes after the start of mix, the assembly shall be removed from the bath. The ring with the impression material shall be separated immediately from the plate. The impression shall be shaken by hand to remove the excess exudate. A gypsum slurry under gentle vibration shall be poured against the impression

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within two minutes from the time the impression is separated from the test block. The gypsum (dental stone), when tested against the block (figure 1)shall reproduce line B (0.050 mm wide). The poured impression shall be placed in an air bath at 23.0 ± 2.0° C and 100 percent relative humidity for 1 hour. The gypsum cast shall be removed and examined. The reproduction of the 0.050 mm line shall be satisfactory if it is continuous for the full width of the ring when viewed under ten power (10X) magnification. Ten minutes after separating the cast from the impression, the 0.075 mm line shall not be obliterated when wiping the forefinger firmly across the surface of the cast at right angles to the inscribed line.

4.4.12 Properties in compression.

4.4.12.1 Preparation of specimens. Test specimens shall be made by placing a ring 3 cm, inside diameter, and 16 mm high, on a flat plate and by filling the ring slightly more than one-half full with alginate material mixed according to the manufacturer's instructions. A mold 12.7 mm, inside diameter, 25.4 mm, outside diameter, and 19 mm high shall be placed immediately inside the ring and shall be forced into the material until the mold touches the plate and the material has exuded onto the top of the mold. A flat plate shall be pressed onto the top of the mold to remove excess material. Two minutes after the start of mix, the mold and its accompanying plates shall be placed in a water bath maintained at 37 ± 1° C. Five minutes and thirty seconds after the start of mix, the mold and plates shall be removed from the water bath. Molds and rings shall be made of polymethyl and methacrylate or metal. Plates shall be made of polymethyl methacrylate, metal or glass.

4.4.12.2 Test conditions. The specimens shall be tested at 23.0 ± 2.0° C. During the test, the specimen shall be protected by a moistened cloth gauze to prevent excessive moisture losses. The gauze shall be wrapped loosely about the specimen so as not to support it.

4.4.12.3 Permanent deformation caused by fixed strain. Six minutes from the start of mix, a test specimen, prepared as specified in 4.4.12.1 and 4.4.12.2, shall be placed in a suitable instrument consisting essentially of a dial gage, sensitive to 0.02 mm, mounted to a steady base and equipped with a screw, positioned in such a manner that sufficient force can be applied to the specimen to produce the required amount of strain (figure 2). Six minutes after the start of the mix, a lightweight plate shall be placed on top of the specimen and the foot of the dial gage allowed to contact the plate. The weight of the plate and the force of the spring in the gage shall be 50 ± 5 g. The dial gage shall be read 30 seconds after the foot of the gage contacts the plate and the value recorded as reading A. The dial gage shall be lowered 2.3 mm by means of the screw and the pressure shall be maintained on the specimen for 30 seconds, then released and the specimen allowed to rest under no load (except that of the

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lightweight plate) for 30 seconds. At end of the 30-second rest period, the dial gage shall be lowered onto the plate and a second reading taken after 30 seconds and the value recorded as reading B. The difference between readings A and B, divided by the original length of the specimen and multiplied by 100 shall be recorded as the percent permanent deformation. The average permanent deformation of three specimens shall be reported.

4.4.12.4 Compressive strength. Eight minutes from the start of mix, a specimen prepared as specified in 4.4.12.1 shall be placed in a suitable machine, accurate to 50 g, and tested for compressive strength. A piece of bond paper shall be placed under and over the specimen in the machine. The specimens shall be loaded continuously and as uniformly as possible to give an average rate of 10 ± 2 kilograms (kg) per minute until fracture. The maximum load at fracture shall be recorded to the nearest 50 g. The maximum load shall be divided by the cross-section area of the mold and reported in g per sq cm. The value for the three determinations (specimens) shall be averaged and reported as the compressive strength.

4.4.12.5 Strain in compression. Ten minutes from the start of mix, the specimen prepared as specified in 4.4.12.1 shall be transferred to an appropriate instrument (figure 3) and shall be subjected to a load calculated to produce a stress of 100 g per sq cm. Thirty seconds later the dial indicator shall be read. This value shall be designated as reading A. Sixty seconds after application of a stress of 100 g per sq cm, an additional load calculated to produce a total stress on the specimen of 1000 g per sq cm, shall be gradually applied during an interval of 10 seconds. Thirty seconds after initiation of the stress of 1000 g per sq cm, a reading of the dial indicator shall be taken. This reading shall be designated as reading B. The difference between readings A and B divided by the original length of the specimen, (the original length of the specimen shall be considered as the height of the mold used in forming it), times 100, shall be considered as the percentage of strain between the stresses of 100 and 1000 g per sq cm. The test shall be performed on three specimens and the values averaged.

4.4.12.6 Deterioration. One unopened unit container of each lot or control number or a sufficient quantity of the powder in a suitable sized closed container such as a jar with a screw cap lid shall be stored for one week at 60 ± 1° C and 100 percent relative humidity. At the end of the storage period, three specimens shall be made and tested in accordance with 4.4.12.4, and three specimens shall be made and tested in accordance with 4.4.9. The value for the three determinations (specimens) shall be averaged and reported as the compressive strength and setting time, respectively, after storage. The results after storage shall be compared with the average of the results for all the original determinations for the lot or control involved to determine acceptance or rejection.

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4.4.13 Consistency. A metal tray commonly used for hydrocolloid impression materials shall be filled with a mix of the alginate. At the end of the mixing and loading time as established by the manufacturer's instruction, the filled tray shall then be inverted for an interval of 15 seconds and observed for any evidence of impression material coming out of tray.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One can, containing the quantity of impression material specified in 3.2.1, together with powder and water proportioners, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be packaged in a can conforming to PPP-C-96, type III, except a commercial coating shall be applied to the exterior of the can. Metallic coating shall be 1.25 hot-dipped or 0.25 electrolytic. Interior coatings shall be as specified in PPP-C-96. The leakage test, as specified in PPP-C-96, shall apply.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty-four (24) units of one type only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant. Closure and strapping shall be in accordance with the appendix of the applicable container specifications.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproof in accordance with appendix of PPP-B-636.

5.2.2 Level B. Twenty-four (24) units of one type only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Military activities.

5.3.1.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract number, lot (control) number, and name of contractor shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F)". Marking shall be accomplished by lithograph process. Pressure sensitive plastic or plastic coated labels may be used. Type II shelf life marking shall be shown as specified in MIL-STD-129.

5.3.1.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Type II shelf life marking shall be shown as specified in MIL-STD-129. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F)".

5.3.2 Civil agencies. In addition to any special marking specified in the contract or order, each unit package and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) To whom statement on irritation is to be furnished (see 3.1.2).

6.2 Artificial dental stone. Dental stone referenced in this specification shall be in accordance with U-S-746.

6.3 Conversion factors.

25.4 mm (2.54 cm) = 1 inch
(^{o}C x 1.8) + 32° = ^{o}F
453.59g = 1 pound
9.807N = 1 kilogram force
9.807 MN/m^2 = 1 Kg/m^2

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sive moisture losses. The gauze shall be wrapped loosely about to prevent exces

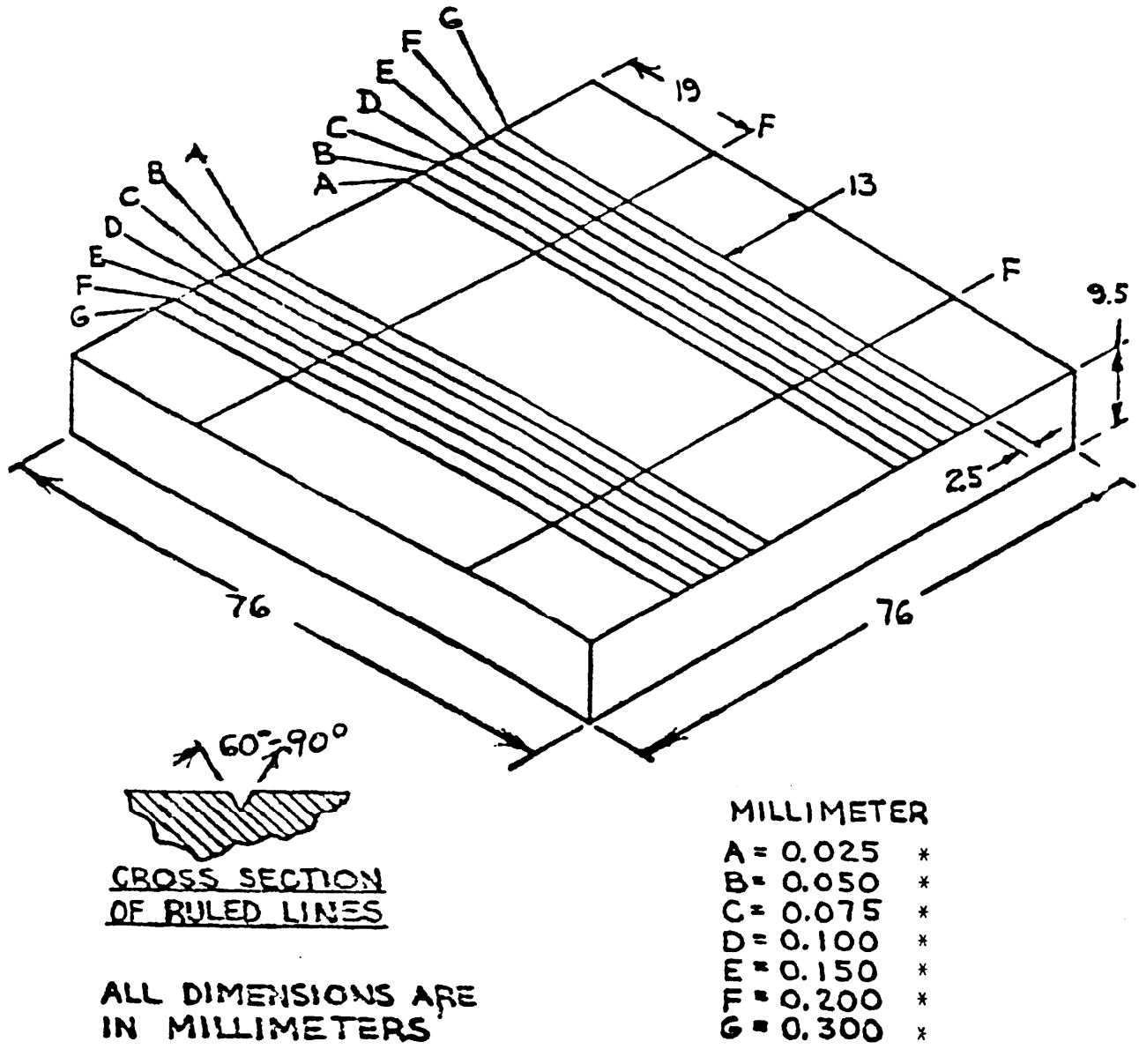


FIGURE 1. Test block for detail reproduction

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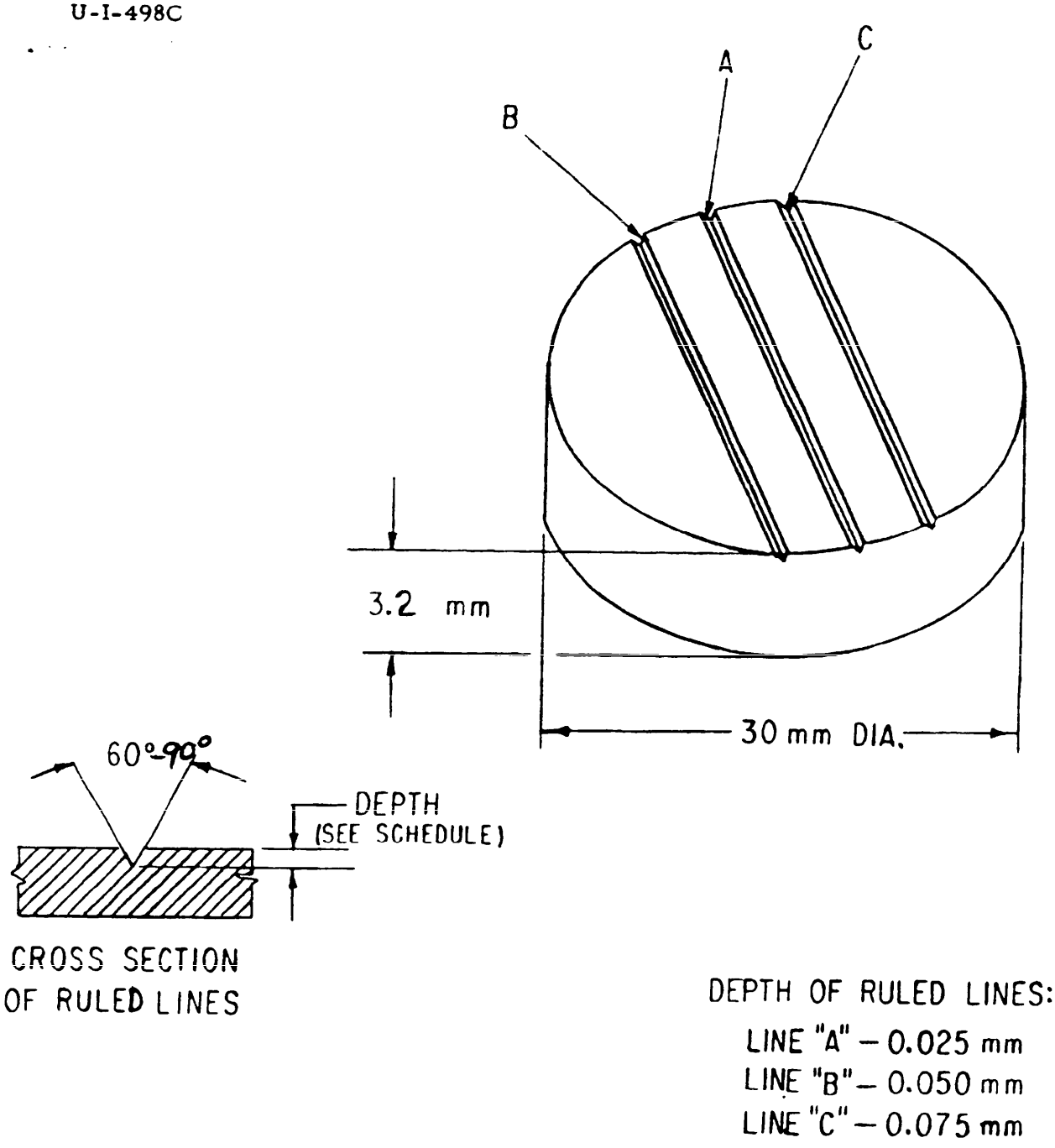


FIGURE 2. Alternate test block for detail reproduction

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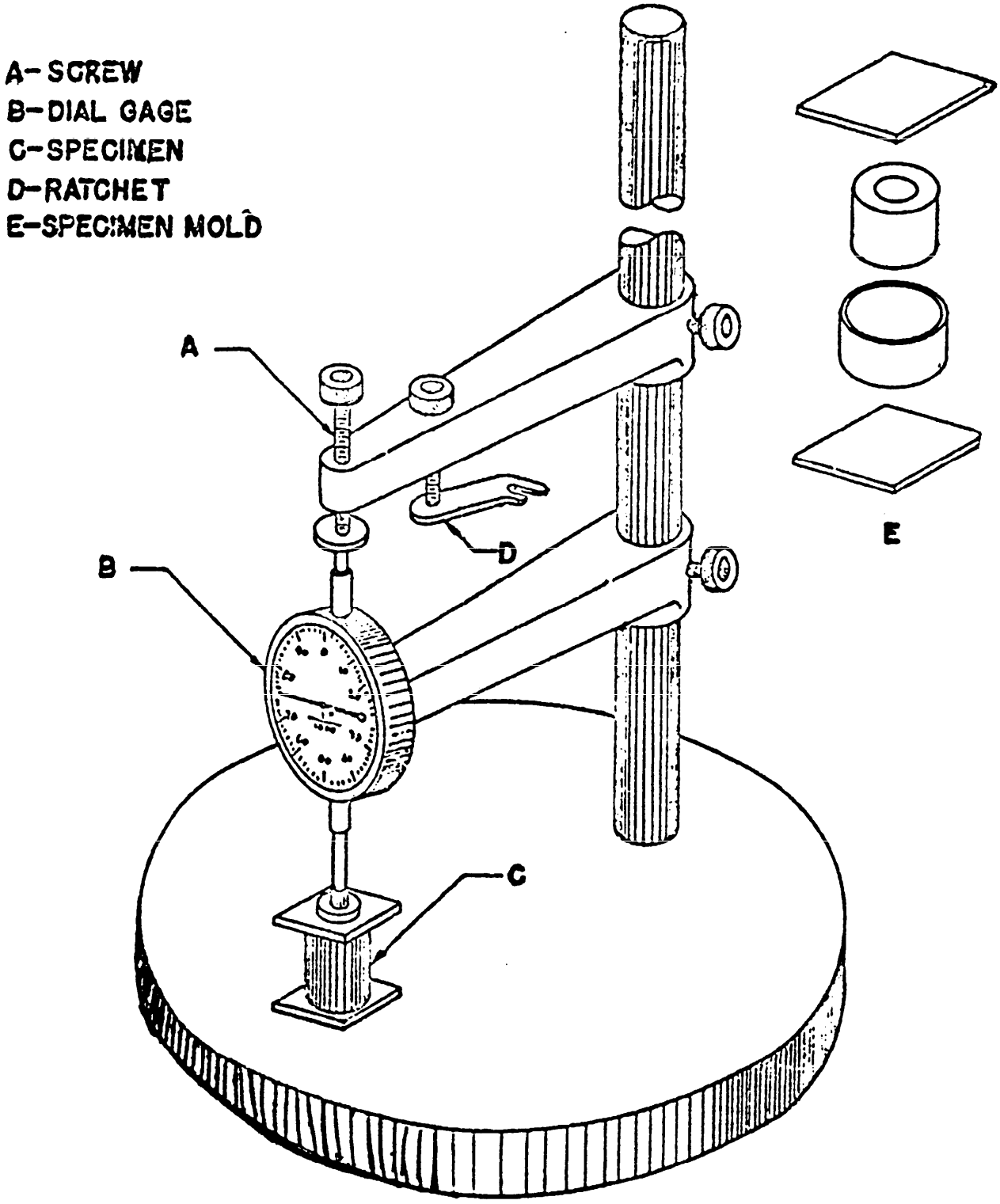


FIGURE 3.

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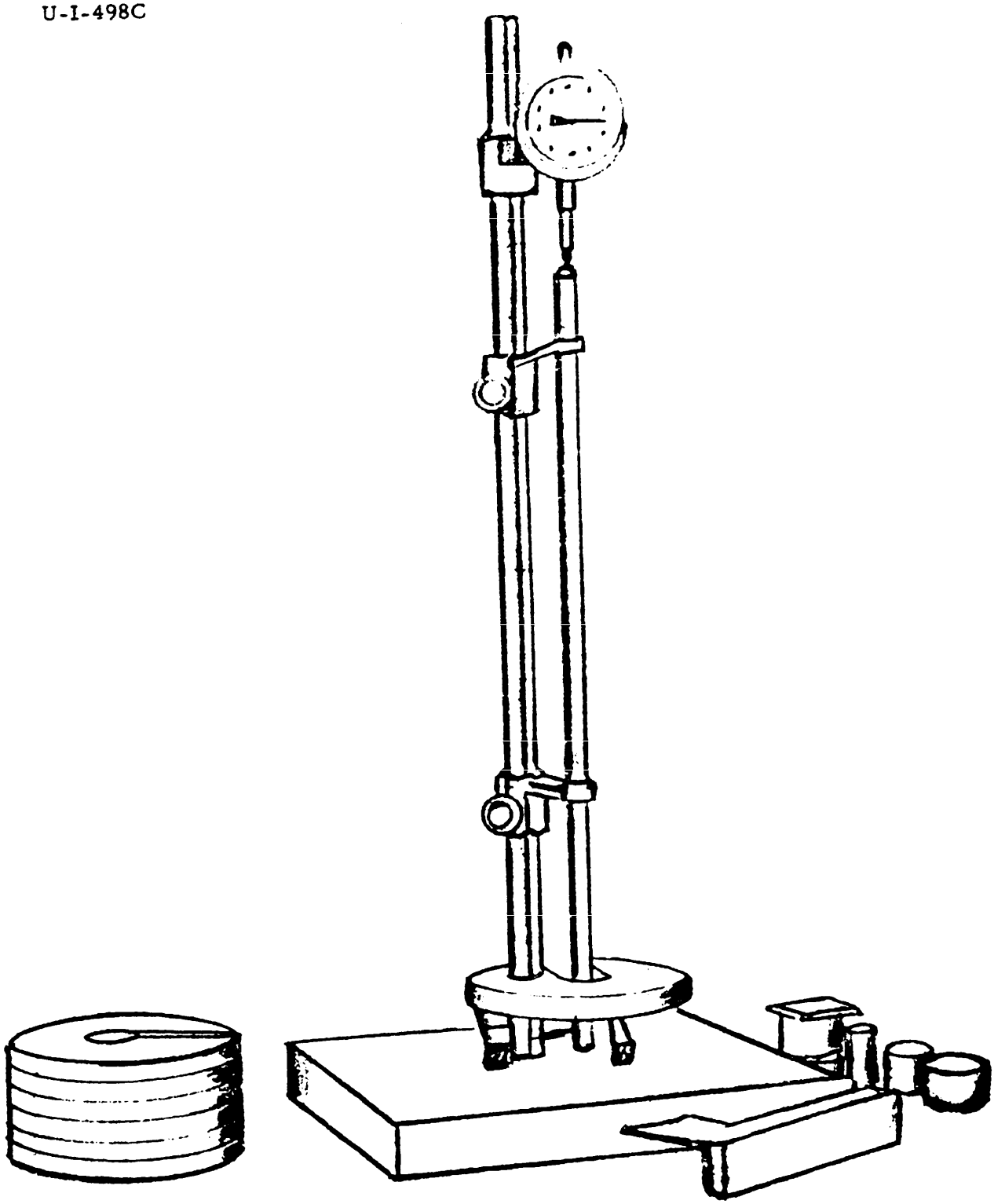


FIGURE 4. Fixture for determining properties in compression

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DOCUMENT IDENTIFIER AND TITLE

U-I-498C IMPRESSION MATERIAL, DENTAL HYDROCOLLOID, ALGINATE TYPE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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